especially in comparison to the complexity of correctly distinguishing between GEDCOM 5.5 and 5.5.1 files. A system will call upon its GEDCOM 5.5.5 reader after detecting a HEAD.GEDC.VERS value of `5.5.5`.
When called upon, the GEDCOM 5.5.5 reader starts reading the GEDCOM file from the beginning, starting with the GEDCOM header. If the header is not completely valid, the file *must* be rejected as *not GEDCOM*, specifically as not even having a correct GEDCOM header.

GEDCOM 5.5.1

Many GEDCOM readers for GEDCOM 5.5.1 and earlier are quite forgiving about GEDCOM errors including errors in the GEDCOM header. This is a very undesirable bad practice, that effectively licenses the creators of sloppy GEDCOM with bad GEDCOM headers to continue doing so. The existing GEDCOM readers for GEDCOM 5.5.1 and earlier should continue their current practices for backward compatibility's sake, so as to not break existing user practices.

All developers are encouraged to fix problems with the GEDCOM files produced by their existing systems. That starts by making sure the system produces a correct and valid GEDCOM header.
Developers are not merely encouraged to take advantage of GEDCOM validators to verify the GEDCOM produced by their systems, but strongly encouraged to make validator usage an integral part of their software development cycle.

Developers of systems that do not support GEDCOM 5.5.5 yet, should make sure that their systems recognise GEDCOM 5.5.5, admits the implementation limitation with a message such as “GEDCOM 5.5.5 not supported yet”, and then aborts processing.

Developers are encouraged to add GEDCOM 5.5.5 support as soon as possible.

GEDCOM 5.5.1 Version Detection

GEDCOM 5.5.1 version detection is complicated by the fact that several applications produce GEDCOM 5.5.1 files that incorrectly claim to be version `5.5` files instead of `5.5.1` files. In fact, nowadays (2019 CE), most GEDCOM files that claim to be GEDCOM 5.5 files are actually GEDCOM 5.5.1 files.

FamilySearch PAF is the best known application to lie about the GEDCOM version used, but not the only one. Other products that use GEDCOM 5.5.1 but lie that they are using GEDCOM 5.5 include Ancestral Quest, MyHeritage Family Tree Builder, Ancestry.com's New Family Tree Maker, Family Tree Heritage, Geni.com, GEDitCOM II, gramps, Heredis 98, Kith and Kin, Legacy Family Tree, Ancestry.com's Online Family Tree, Personal Ancestry Writer II, Reunion, The Next Generation of Genealogy Sitebuilding (TNG), and WikiTree.

Some products that correctly identify their use of GEDCOM 5.5.1 include Ahnenblatt, Family Echo, Ancestry.com's New Family Tree Maker, findmypast Family Tree, gedcom4j, Legacy Family Tree, MacFamilyTree, Picturae's Memorix Maior, phpGedView, RootsMagic, and WebTrees.

Some products appear in both list because some versions of the product get it wrong, while later versions get it right. For the detection algorithm presented here, all that matters is when products started using GEDCOM 5.5.1 yet continued to label their GEDCOM files (incorrectly) as GEDCOM 5.5. It does not matter whether or when the developers fixed their product to start labelling their GEDCOM 5.5.1 files correctly.

GEDCOM 5.5 versus GEDCOM 5.5.1 File Content

It is possible to distinguish between GEDCOM 5.5 and GEDCOM 5.5.1 files based on their content, for example with the following rules:

- if the GEDCOM files contains a OBJE.BLOB record, it is GEDCOM 5.5
- if the GEDCOM files contains a PLAC.MAP record, it is GEDCOM 5.5.1

However, a GEDCOM reader should be able to determine the actual GEDCOM version without looking beyond the GEDCOM header.

GEDCOM 5.5.1 Detection Algorithm

This is a the algorithm for distinguishing between GEDCOM 5.5 and GEDCOM 5.5.1. Numbered steps should be performed in the order shown. As soon as a rule matches and the actual GEDCOM version is known, no further tests should be performed.

- Detect the character encoding first
- read the HEAD.GEDC.VERS line value
- if the value is `5.5.1`, it is GEDCOM 5.5.1
- if the value is `5.5`, it may be either GEDCOM 5.5 or GEDCOM 5.5.1

1. (Optional): check the system identifier (HEAD.SOUR line value)
   against a list of system identifiers for products discontinued before (or some time after) the introduction of GEDCOM 5.5.1:
   if there is a match, it's definitely GEDCOM 5.5, and not GEDCOM 5.5.1.

2. Check the GEDCOM encoding
   - if the character encoding is UTF-8, it is GEDCOM 5.5.1

3. Check for tags introduced in GEDCOM 5.5.1
   - if the header contains a HEAD.SOUR.CORP.ADR3 record, it is GEDCOM 5.5.1
   - if the header contains a HEAD.SOUR.CORP.EMAIL record, it is GEDCOM 5.5.1

- if the header contains a HEAD.SOUR.CORP.FAX record, it is GEDCOM 5.5.1
- if the header contains a HEAD.SOUR.CORP.WWW record, it is GEDCOM 5.5.1

4. Check for known products and versions.

    1. Make sure you have a system identifier.
        - if the system identifier is too long (it is for The Next Generation of Genealogy Sitebuilding), issue a non-fatal error and continue
        - If the (HEAD.SOUR line value is missing (illegal!)
            - try to identify the correct system identifier by other means (see section on MacFamilyTree below)
            - if you failed to identify it, issue a fatal error and abort
            - if you managed to identify it, issue a non-fatal error and continue with that system identifier

    2. Make sure you have a version number.
        - If the (HEAD.VERS line value is missing, issue a fatal error and abort.
        - If the (HEAD.VERS line value isn't a proper version number
            - try to extract the version number (see section on Family Tree Maker below)
            - if you failed to extract it, issue a fatal error and abort
            - if you managed extract it, issue a non-fatal error and continue with that version number

    3. Perform the known product / version number checks.
        - if the system identifier is `PAF`
            - if the version number is less than `5.0`, it is GEDCOM 5.5
            - if the version number is `5.0` or more, it is GEDCOM 5.5.1
        - See the table below for more products and version numbers.

5. Check for user-defined tags being used instead of GEDCOM 5.5.1 tags
    - if the header contains a HEAD.SOUR.CORP._EMAIL record, it must be GEDCOM 5.5
    - if the header contains a HEAD.SOUR.CORP._FAX record, it must be GEDCOM 5.5
    - if the header contains a HEAD.SOUR.CORP._WWW record, it must be GEDCOM 5.5

6. None of the previous rules matched? Assume it is 5.5 as stated.

- if the HEAD.GEDC.VERS value is `5.5`, but the actual GEDCOM version is 5.5.1, issue a non-fatal error
- Issue an informative message that the file will be processed as GEDCOM 5.5 or GEDCOM 5.5.1, whichever version was detected

Notice the last two instructions in the algorithm. It is good to let the user know that an ostensible GEDCOM 5.5 file is really a GEDCOM 5.5.1 file. It is good to explicitly inform the user how the file will be processed.

Product Table

Version of product that started using GEDCOM 5.5.1		
application	**system identifier**	**version**
Personal Ancestral File	PAF	5.0
Ancestral Quest	AncestQuest	12.0
Family Tree Maker	FTM	21.0.0.466
GenoPro	GenoPro	2.0
gramps	Gramps	2.0
Lifelines	Lifelines	3.0
MacFamilyTree	MacFamilyTree	5.7.8
MagiKey Family Tree	MagiKey Family Tree	*all versions*
Family Tree Builder	MYHERITAGE	5.5
Reunion	Reunion	9.0
RootsMagic	RootsMagic	*all versions*
The Next Generation of Genealogy Sitebuilding	The Next Generation of Genealogy Sitebuilding	7.0
PRO-GEN	PRO-GEN	3.0
This list is not exhaustive.		

Product Information

Family Tree Maker Version Number

The Family Tree Maker name is used for what are really three different, consecutive products (and then there's also Family Tree Maker for Macintosh):

- the original Family Tree Maker, an MS-DOS application
- Family Tree Maker for Windows a Windows applications
- New Family Tree Maker for Windows, a Microsoft .NET application

GEDCOM System Identifiers

Those three different products use only two different GEDCOM system identifiers. Family Tree Maker for MS-DOS uses `FTM` Family Tree Maker for Windows uses `FTW` and New Family Tree Maker for Windows uses `FTM` again.
This is wrong, and may mess up some tests, but does not mess up this test, because the version numbers continued to increase with new products. Family Tree Maker 2012 Service Pack 2 (version 21.0.0.46) is the first version to produce ostensible GEDCOM 5.5 files that are really GEDCOM 5.5.1 files. Family Tree Maker for MS-DOS pre-dates GEDCOM 5.5.1, and all its versions numbers are considerably lower than those of New Family Tree Maker for Windows. So, if the Family Tree Make version number is 21.0.0.466 or later, it is GEDCOM 5.5.1.

FTM Version Number

When Ancestry.com created New Family Tree Maker for Windows, they messed up the version number record, by practically treating it as product name record. This was only fixed after Software MacKiev became the owner of Family Tree Maker.
Family Tree Maker 2012's version number is `21.0.0.38`, but the HEAD.SOUR.VERS line value isn't `21.0.0.38`, it is `Family Tree Maker (21.0.0.38)` instead. That value is illegal, for two reasons:

- it clearly isn't a proper version number
- this value (29 characters) is longer than the maximum HEAD.SOUR.VERS line value length

of 15 code units

A GEDCOM reader should definitely issue an error when encountering this, and it is okay for that error to be an fatal error; it is fine to reject an ostensible GEDCOM file if it does not even have a valid GEDCOM header.
It is not difficult to recognise the `Family Tree Maker (version number)` pattern and extract the version number from it, and this is what a GEDCOM reader has to do before it can perform this test.

32-bit versus 64-bit Version Number

From Family Tree Maker 2014 onwards, Family Tree Maker comes in both a 32-bit and a 64-bit build, and those two different builds have different version numbers. The version number of the 32-bit Family Tree Maker 2014 is `22.0.0.207`, while the version number of the 64-bit Family Tree Maker 2014 is `22.0.0.1207`.
This does not complicate the test, as the switch to GEDCOM 5.5.1 happened with version Family Tree Maker 2012 Service Pack 2 (version 21.0.0.46), for which there is only a 32-bit build, and therefore just one version number to compare to.

Family Tree Maker for Macintosh

There are two Family Tree Maker for Mac products.

- Family Tree Maker for Mac, based on Family Tree Maker for Windows, released by Broderbund in 1997 CE
- Family Tree Maker for Mac, based on New Family Tree Maker, released by Ancestry.com in 2010

The new Family Tree Maker for Mac uses the same system identifier (`FTM`) as New Family Tree Maker (for Windows), and that is wrong, because it is definitely another product. Family Tree Maker for Mac uses the same version numbering as New Family Tree Maker, and the older Ancestry.com Family Tree Maker for Mac also uses the same illegal HEAD.SOUR.VERS format. The GEDCOM 5.5.1 detection code for Family Tree Maker does not to distinguish between Family Tree Maker (for Windows) and Family Tree Maker for Mac.

Recognising MacFamilyTree GEDCOM files

Recognising MacFamilyTree GEDCOM files is harder than it should be. In many MacFamilyTree GEDCOM files, the HEAD.SOUR value is empty. This is illegal, and a GEDCOM reader should definitely issue an error, and it is okay for that error to be an fatal error, because it's fine to reject an ostensible GEDCOM file if it does not even have a valid GEDCOM header.
It is possible for a GEDCOM reader to recognise the GEDCOM file as MacFamilyTree GEDCOM file; if the HEAD.SOUR line value is empty, but the HEAD.SOUR.NAME line value is `MacFamilyTree`, it is a MacFamilyTree GEDCOM file.

Legacy Family Tree

Legacy Family Tree GEDCOM files may lack the GEDCOM version number! See the *Legacy Family Tree GEDCOM Version* annotation.

Table Updates

This algorithm can be improved by updating the table with more product-specific information. This bonus chapter is based on and practically identical to the *GEDCOM 5.5.1 Version Detection* article, which is likely to be updated sooner and more frequently than this chapter.

References

- GEDCOM 5.5.1 Version Detection
- GEDCOM Version Detection
- Truncated GEDCOM Version
- GEDCOM System Identifiers
- New Family Tree Maker versions
- Ancestry.com & Software MacKiev Family Tree Maker GEDCOM Header

Event GEDCOM

Back in 1994, CommSoft, the creators of the Roots series of genealogy applications, defined an alternative GEDCOM form. It was originally called InterGED, but best known as Event GEDCOM. Event GEDCOM was first supported by ROOTS IV.
Although Event GEDCOM is arguably superior to the Lineage-Linked Form, Event GEDCOM never caught on. Event GEDCOM was only ever supported by Roots IV, Roots V, Family Gathering and Ultimate Family Tree.
Event GEDCOM files are rare, and backward compatible with GEDCOM 5.3 lineage-linked GEDCOM files.

FamilySearch Misinformation

FamilySearch's statement about CommSoft's Event GEDCOM in GEDCOM 5.4, 5.5, 5.5.1 (and even the 5.6 draft) is incorrect. Event GEDCOM *does not* use the line value "`EVENT_LINEAGE_LINKED`", neither with nor without quotes. Event GEDCOM files are backward compatible with lineage-linked GEDCOM files, and therefore use the line value `LINEAGE-LINKED`, *as instructed by the FamilySearch GEDCOM specification*: “Systems will use this value to specify GEDCOM compatible with these specifications.”.

Actual Event GEDCOM

Event GEDCOM 1.0 files contain a additional subrecord, HEAD.FORM.FORM record with the value `EVENT` (illegal in LINEAGE-LINKED GEDCOM), and, although based on GEDCOM 5.3, claims to be GEDCOM version 1.0.

```
0 HEAD
...
1 SOUR FAMGATH
2 VERS 1.0
2 NAME Family Gathering
2 CORP Palladium Interactive, Inc.
...
1 GEDC
2 VERS 1.0
2 FORM LINEAGE-LINKED
3 FORM EVENT
1 DEST FAMGATH
...
```

Family Gathering 1.0 Event GEDCOM header.

References

- GEDCOM Form
- Event GEDCOM Detection

Known GEDCOM Form Errors

Many GEDCOM 5.5 and 5.5.1 readers correct for known GEDCOM 5.5 and 5.5.1 GEDCOM `LINEAGE-LINKED` spelling errors. A GEDCOM 5.5.5 reader *need not* and *must not* correct for spelling errors.

If a GEDCOM 5.5.5 reader encounters an unknown GEDCOM form, it *must not* continue processing, but must abort with an *unsupported GEDCOM form* message.

`LINEAGE_LINKED` instead of `LINEAGE-LINKED` (underscore instead of dash)

Millennia's Legacy Family Tree (now owned by MyHeritage) is perhaps the best known but not the only genealogy application to get this wrong. This error occurs in GEDCOM files produced by EasyTree, FamilyMatters, GEDitCOM, Généamania, GenoPro, Secere Family Tree and WebSSG.

`LINAGE-LINKED` instead of `LINEAGE-LINKED` (first E missing):

This error occurs in GEDCOM files created by GenoPro.

`Lineage - Linked` instead of `LINEAGE-LINKED` (spaces before and after the dash):

The French application Genealogic by Infoduc and the French site Genealogie.com created GEDCOM files containing this error.

More than One <<SUBMITTER_RECORD>>

A GEDCOM 5.5.5 file contains exactly one <<SUBMITTER_RECORD>>, directly after the <<HEADER>>

A typical GEDCOM 5.5.1 file contains exactly one <<SUBMITTER_RECORD>>, referenced by HEAD.SUBM, but GEDCOM 5.5.1 allowed multiple submitter records.

Both the <<INDIVIDUAL_RECORD>> and the <<FAM_GROUP_RECORD>> optionally reference a submitter record. This allows multiple submitters to be associated with different records.

This GEDCOM 5.5.1 feature remained widely unsupported. It makes little sense for single-user applications, and even collaborative applications (social genealogy sites) do not use it. All genealogy applications, including FamilySearch PAF support only one submitter per GEDCOM file.
The editor and reviewers are not aware of any genealogy application that supports more than submitter per GEDCOM file. Even if there is one, other genealogy applications are not able to import and process that data. The GEDCOM 5.5.1 Annotated Edition strongly deprecated this feature. The GEDCOM 5.5.5 specification demands exactly one submitter record.

GEDCOM Writer Best Practice

- use GEDCOM 5.5.5 or later

GEDCOM 5.5.1 Writer Best Practice

- create exactly one submitter record
- have HEAD.SUBM reference this submitter record
- write the submitter record immediately after the GEDCOM header

GEDCOM 5.5.1 Reader Best Practice

- Remember the HEAD.SUBM cross-reference identifier
- Keep track of submitter record identifiers like you keep track of other identifiers
  - if any cross-reference identifier is reused, issue a fatal error (identifier already used) and abort the import
  - if the referenced submitter record is not found, issue a fatal error (referenced record not found) and abort the import

- Until you have the right submitter record:
  - Accept each submitter record you encounter (overwriting the previous one, if any)
  - issue an error if it isn't the one referenced by HEAD.SUBM

- Once you have the right submitter record, for each additional submitter record:
  - do *not* overwrite the submitter record you already found
  - issue an *implementation limitation* error (only one submitter record supported) and move on

No Standard for Multimedia File Transfer

The GEDCOM specification defines a <<MULTIMEDIA_RECORD>>. Each multimedia record provides a link to an external file. The GEDCOM specification does not provide a standard for bundling multimedia files with a GEDCOM file.

As long as the source and destination system are two different applications on the same computer (or mirrored systems), the multimedia file links are efficient, but for sharing multimedia files to another system, a locally valid link is inadequate.
Links to web resources do remain valid between different systems, but their permanence cannot be guaranteed, and performance is likely to suffer.

The FamilySearch GEDCOM specification does not provide any standard, any rules or guidelines for bundling multimedia files with a GEDCOM file. It is not uncommon to bundle a GEDCOM file and media together in a ZIP file, but that only takes care of the media transfer, not the file paths, which are almost sure to be different on different systems, especially if these are managed by different users.

Some desktop applications have built-in smarts to try and repair files references, and several desktop applications provide syncing of desktop databases with web databases.

Best Practices

Applications should maintain links to media, *not* store media inside their database. Storing media inside the database bloats the database, and makes the media inaccessible to other tools.
Genealogy software developers and applications should encourage users to collect and keep all related media in one directory (with subdirectories).

GEDCOM 5.5 <<MULTIMEDIA_RECORD>>

The FamilySearch GEDCOM 5.5.1 <<MULTIMEDIA_RECORD>> is different from the GEDCOM 5.5 <<MULTIMEDIA_RECORD>>.

This is the GEDCOM 5.5.1 <<MULTIMEDIA_RECORD>> (corrected to use <MULTIMEDIA_FILE_REFERENCE> instead of the non-existing <MULTIMEDIA_FILE_REFN>):

MULTIMEDIA_RECORD:=

```
n @<XREF:OBJE>@ OBJE                                  {1:1}
  +1 FILE <MULTIMEDIA_FILE_REFERENCE>                 {1:1}
    +2 FORM <MULTIMEDIA_FORMAT>                       {1:1}
      +3 TYPE <SOURCE_MEDIA_TYPE>                     {0:1}
    +2 TITL <DESCRIPTIVE_TITLE>                       {0:1}
  +1 REFN <USER_REFERENCE_NUMBER>                     {0:M}
    +2 TYPE <USER_REFERENCE_TYPE>                     {0:1}
  +1 RIN <AUTOMATED_RECORD_ID>                        {0:1}
  +1 <<NOTE_STRUCTURE>>                               {0:M}
  +1 <<SOURCE_CITATION>>                              {0:M}
  +1 <<CHANGE_DATE>>                                  {0:1}
```

The BLOB context of the multimedia record was removed in version 5.5.1. A reference to a multimedia file was added to the record structure. The file reference occurs one to many times so that multiple files can be grouped together, each pertaining to the same context. For example, if you wanted to associate a sound clip and a photo, you would reference each multimedia file and indicate the format using the FORM tag subordinate to each file reference.

The <<SOURCE_CITATION>> line is not new in GEDCOM 5.5.1.
It was added to GEDCOM 5.5 through the GEDCOM 5.5 errata sheet.

The GEDCOM 5.5.1 <<MULTIMEDIA_RECORD>> allows a single OBJE record to link to multiple related files.
This feature is not widely supported; many genealogy application will read only the first OBJE.FILE. To avoid loss of data, it is recommended to have just one OBJE.FILE subrecord per OBJE record. The GEDCOM 5.5.5 specification simplified the multimedia record: one multimedia record per file, one file per multimedia record.

This is the GEDCOM 5.5 <<MULTIMEDIA_RECORD>> for comparison:

MULTIMEDIA_RECORD:=

```
n @<XREF:OBJE>@ OBJE                                  {1:1}
  +1 FORM <MULTIMEDIA_FORMAT>                         {1:1}
  +1 TITL <DESCRIPTIVE_TITLE>                         {0:1}
  +1 <<NOTE_STRUCTURE>>                               {0:M}
  +1 <<SOURCE_CITATION>>
  +1 BLOB
    +2 CONT <ENCODED_MULTIMEDIA_LINE>
  +1 OBJE @<XREF:OBJE>@    /* chain to continued object */
  +1 REFN <USER_REFERENCE_NUMBER>                     {0:M}
    +2 TYPE <USER_REFERENCE_TYPE>                     {0:1}
  +1 RIN <AUTOMATED_RECORD_ID>                        {0:1}
  +1 <<CHANGE_DATE>>                                  {0:1}
```

Large whole multimedia objects embedded in a GEDCOM file would break some systems. For this purpose, large multimedia files should be divided into smaller multimedia records by using the subordinate OBJE tag to chain to the next <MULTIMEDIA_RECORD>

fragment. This will allow GEDCOM records to be maintained below the 32K limit for use in systems with limited resources.

Within the definition above, the BLOB subrecord has been marked as deprecated (in GEDCOM 5.5), because on second thoughts, FamilySearch considered BLOBs a bad idea, so they discontinued BLOB support in GEDCOM 5.5.1.
However, it is best to think of this entire GEDCOM 5.5 record as obsolete. Developers should pay attention to the small differences between the GEDCOM 5.5 and 5.5.1 definition, and not simply reuse old code, but make sure that they are using the GEDCOM 5.5.1 definition.

Chain Pointer

The <XREF:OBJE> pointer within the BLOB subrecord (on the line highlighted because it includes an illegal C-style comment) isn't a circular reference. As the GEDCOM 5.5 definition explains, it's there because encoded media file are likely to be larger than the (now obsolete) maximum top-level record size of 32 KB. A GEDCOM 5.5 writer must break the encoded media files into pieces of less than 32KB, and then chain those pieces together. A GEDCOM 5.5 reader must follow the chain pointers and put the encoded media file together again.

GEDCOM 5.5.1 <MULTIMEDIA_LINK>

This is the GEDCOM 5.5.5 <<MULTIMEDIA_LINK>> definition: {1:M}

MULTIMEDIA_LINK:=

n **OBJE** <XREF:OBJE> {1:1}

A <<MULTIMEDIA_LINK>> is a link to a <<MULTIMEDIA_RECORD>> (which is a link to a multimedia file). {0:1}
There should be one multi-media record per file. There may be multiple multi-media links to a multi-media record. {0:1}

That GEDCOM 5.5.5 <<MULTIMEDIA_LINK>> is a significant simplification of the GEDCOM 5.5.1 definition (without page numbers, but with annotations & corrections from the *Annotated Edition*):

MULTIMEDIA_LINK:=

[
n **OBJE** @<XREF:OBJE>@ {1:1}
|
n **OBJE**
~~+1 **FILE** <MULTIMEDIA_FILE_REFN>~~
+1 **FILE** <MULTIMEDIA_FILE_REFERENCE>
+2 **FORM** <MULTIMEDIA_FORMAT>
+3 MEDI <SOURCE_MEDIA_TYPE>
+1 TITL <DESCRIPTIVE_TITLE>
]

Note: some systems may have output the following 5.5 structure. The new context above was introduced in order to allow a grouping of related multimedia files to a particular context.

n OBJE
+1 FILE <MULTIMEDIA_FILE_REFERENCE>
+1 FORM <MULTIMEDIA_FORMAT>
~~+2 MEDI <SOURCE_MEDIA_TYPE>~~
+1 TITLE <DESCRIPTIVE_TITLE>
+1 <<NOTE_STRUCTURE>>

Non-Existent <MULTIMEDIA_FILE_REFN>

The <<MULTIMEDIA_LINK>> syntax references <MULTIMEDIA_FILE_REFN>, but the GEDCOM lineage-linked form doesn't define <MULTIMEDIA_FILE_REFN>, it defines <MULTIMEDIA_FILE_REFERENCE>.

This is the *actual* GEDCOM 5.5 <MULTIMEDIA_LINK> definition:

MULTIMEDIA_LINK:=

[/* embedded form */
n **OBJE** @<XREF:OBJE>@ {1:1}
| /* linked form */

```
n OBJE
   +1 FORM <MULTIMEDIA_FORMAT>                                  {1:1}
   +1 TITL <DESCRIPTIVE_TITLE>                                  {0:1}
   +1 FILE <MULTIMEDIA_FILE_REFERENCE>                          {1:M}
   +1 <<NOTE_STRUCTURE>>                                        {0:M}
]
```

This structure provides two options in handling the GEDCOM multimedia interface. The first alternative (embedded) includes all of the data, including the multimedia object, within the transmission file. The embedded method includes pointers to GEDCOM records that contain encoded image or sound objects. Each record represents a multimedia object or object fragment. An object fragment is created by breaking the multimedia files into several multimedia object records of 32K or less. These fragments are tied together by chaining from one multimedia object fragment to the next in sequence. This procedure will help manage the size of a multimedia GEDCOM record so that existing systems which are not expecting large multimedia records may discard the records without crashing due to the size of the record. Systems which handle embedded multimedia can reconstitute the multimedia fragments by decoding the object fragments and concatenating them to the assigned multimedia file.

The second method allows the GEDCOM context to be connected to an external multimedia file. This process is only managed by GEDCOM in the sense that the appropriate file name is included in the GEDCOM file in context, but the maintenance and transfer of the multimedia files are external to GEDCOM.

Two <<MULTIMEDIA_LINK>> Forms

The GEDCOM 5.5 and GEDCOM 5.5.1 <<MULTIMEDIA_LINK>> definition allows two alternative forms The first form, called the *embedded form*, uses an <XREF:OBJE>, a link to a <<MULTIMEDIA_RECORD>>, while the second form, called the *linked form*, contains details for an external multimedia file.

The GEDCOM 5.5 <<MULTIMEDIA_LINK>> definition makes it clear that the so-called *embedded form* was meant for embedded multimedia objects only, and GEDCOM 5.5.1 does not support embedded multimedia objects.
The *GEDCOM 5.5.1. Annotated Edition* notes that you might think that this implies that the *embedded form* is obsolete now, but that it is not; on the contrary, it is the preferred form.

The so-called *linked form* of the GEDCOM 5.5.1 <<MULTIMEDIA_LINK>> record offers a subset of the GEDCOM 5.5.1 <<MULTIMEDIA_RECORD>> features. That makes it superfluous, and that is why the *Annotated Edition* deprecated the *linked form*.

The *GEDCOM 5.5.1. Annotated Edition* provides the following best practice for GEDCOM 5.5.1:

> It is recommended that all multimedia files be recorded in GEDCOM 5.5.1 using the top-level OBJE record, and that all instances of the <MULTIMEDIA_LINK> record are <XREF:OBJE> links to these <<MULTIMEDIA_RECORD>> records.
> Consider the so-called *linked form* of the <<MULTIMEDIA_LINK>> record deprecated.

GEDCOM 5.5.1 Specification misrepresents GEDCOM 5.5 <<MULTIMEDIA_LINK>>

The GEDCOM 5.5.1 <<MULTIMEDIA_LINK>> definition claims that the (linked form of the) GEDCOM 5.5 <<MULTIMEDIA_LINK>> definition (after the indicated corrections) looks like this:

```
MULTIMEDIA_LINK:=

    n OBJE
      +1 FILE <MULTIMEDIA_FILE_REFERENCE>                    {1:M}
      +1 FORM <MULTIMEDIA_FORMAT>                            {1:1}
        +2 MEDI <SOURCE_MEDIA_TYPE>                          {0:1}
      +1 TITL <DESCRIPTIVE_TITLE>                            {0:1}
```

However, that is not correct. The linked form of the GEDCOM 5.5 <<MULTIMEDIA_LINK>> definition looks like this:

```
MULTIMEDIA_LINK:=

    n OBJE
      +1 FORM <MULTIMEDIA_FORMAT>                            {1:1}
      +1 TITL <DESCRIPTIVE_TITLE>                            {0:1}
      +1 FILE <MULTIMEDIA_FILE_REFERENCE>                    {1:M}
      +1 <<NOTE_STRUCTURE>>                                  {0:M}
```

The GEDCOM 5.5 definition does not include a MEDI subrecord at all, neither as direct subrecord of the OBJE record, nor as a subrecord of the OBJE.FORM record. The GEDCOM 5.5 definition allows notes, while the GEDCOM 5.5.1 definition does not.

Editing Error

The misrepresentation of the GEDCOM 5.5 syntax seems an unintentional editing error. The two differences between the GEDCOM 5.5 and 5.5.1 definitions that the author probably wanted to draw the reader's attention to, are:

- The GEDCOM 5.5 record allows just one file, while the GEDCOM 5.5.1 record allows a group of files
- The GEDCOM 5.5 record has the FORM record as a direct subrecord of the OBJE record, while the GEDCOM 5.5.1 record has the FORM record as a subrecord of the OBJE.FILE record.

Syntax Exception

The real point of the remark is to demand a syntax exception for “some systems” that use GEDCOM 5.5.1, but still create MULTIMEDIA_LINK records in GEDCOM 5.5 format! The implied demand is that GEDCOM 5.5.1 readers should accept GEDCOM 5.5 MULTIMEDIA_LINK records, even though they are not part of GEDCOM 5.5.1.

GEDCOM 5.5.1 specification <<MULTIMEDIA_LINK>> Clairvoyance

The *FamilySearch GEDCOM 5.5.1 specification* states that some systems “may have output” (past tense) the GEDCOM 5.5 structure within GEDCOM 5.5.1 files. That sounds like a helpful annotation, but it is a rather remarkable one.
On the day that FamilySearch introduced GEDCOM 5.5.1, FamilySearch already knew that some applications would use GEDCOM 5.5.1, yet continue to use the GEDCOM 5.5 <MULTIMEDIA_LINK> structure instead of GEDCOM 5.5.1 <MULTIMEDIA_LINK> structure?
Unless the FamilySearch GEDCOM editors own a time machine, something is wrong with their claim, or at least their presentation of it.

PAF GEDCOM

What the GEDCOM 5.5.1 specification really demands, without being open and honest about it, is that third parties are expected to accommodate FamilySearch's unwillingness to follow their own specification in their own product!
FamilySearch's Personal Ancestral File 5.x uses GEDCOM 5.5.1, but still uses the GEDCOM 5.5 <MULTIMEDIA_LINK> structure. So, this FamilySearch statement tells us that, when they were still writing the GEDCOM 5.5.1 specification, FamilySearch had already decided that PAF 5 would use GEDCOM 5.5.1, but keep using the GEDCOM 5.5 <MULTIMEDIA_LINK> structure, and then put this note about “some systems” in to get third parties to support PAF's misbehaviour.

GEDCOM 5.5.1 Writer Best Practice

- Use GEDCOM 5.5.1 records within GEDCOM 5.5.1 files.
- Do not use records from other GEDCOM versions within GEDCOM 5.5.1 (unless they are identical).

GEDCOM 5.5.1 Reader Best Practice

- Because PAF will never be updated again, and still has significant market share, third parties should indeed accept the GEDCOM 5.5 <MULTIMEDIA_LINK> structure within PAF GEDCOM 5.5.1 files - but still issue an error for each occurrence.

GEDCOM 5.5.5 Reader Best Practice

- Demand GEDCOM 5.5.5 records. Reject everything else as a fatal error.

one- and two-digit years illegal

The GEDCOM 5.5.1 specification has this definition of <YEAR>:

> **YEAR:= {Size=3:4}**
>
> A numeric representation of the calendar year in which an event occurred.
> Years 1 through 99 must be padded out to at least 3 digits by using leading zeroes.

Notice that a year is a number that is at least 3 characters long; the FamilySearch GEDCOM specification does not allow years of less than 3 digits. One-digit and two-digit years are illegal.

Three or Four Digits

GEDCOM 3.0 and 4.0 demand that all years be 4 digits long. It was only with GEDCOM 5.0 that FamilySearch started to allow 3-digit dates, even while its *How to record dates* section still instructs the reader to enter a 4-digit date:

> Type the day (one or two characters) first, then the month (capitalize the first three letters of the English name of the month; do not use a period at the end), and then the year (four-character numeric year). The day and month may be omitted if unknown.

The demand that year be at least 3 digits long remains present in all subsequent version of GEDCOM, including GEDCOM 5.6.

No Year less than 100

There is no year zero in the Julian or Gregorian Calendar, the year 1 CE is preceded by the year 1 BCE.
The demand that years must have at least three digits can be interpreted as the FamilySearch GEDCOM specification not allowing the years 1 CE through 99 CE, nor 99 BCE through 1 BCE. This interpretation is certainly encouraged by FamilySearch's *PAF 2.1 Family Records Data Structure Description* (23 Jun 1988), which contains the following definition for an 11-bit year field within a 3-byte date field:

```
YEAR (bits 0-11) - A number between 100 and 2047
```

That's a remarkable definition; it explicitly excludes the years 1 through 99, without giving a reason for doing so.

PAF is based on Ancestral Quest 3.0, and one of the changes listed for Ancestral Quest 3.0 on its version history page is “Allowed for dates to preceed [sic] 100 AD”.

Treat as Date Phrase

The GEDCOM 5.0 specification has this definition of <YEAR>:

> **YEAR:= {Size=3:4, Type=NUMBER}**
>
> A numeric representation of the calendar year in which an event occurred. Years less than 3 digits long will be treated as a number in a phrase.

The second sentence of this definition, “Years less than 3 digits long will be treated as a number in a phrase.”, only occurs in GEDCOM 5.0. It's gone in GEDCOM 5.3, which suggests that

FamilySearch reconsidered, and did not think that particular instruction a good idea.

Still, this definition makes it crystal clear that FamilySearch does not like years of two digits or less, that the minimum length of 3 characters is quite deliberate. FamilySearch does not explain this limitation, nor what to do with years smaller than 100.

This restriction does not exist to combat the tendency of people to abbreviate dates, by leaving off the century. Such a genealogy would be so confusing to read, that the author would quickly start using full years. Moreover, genealogy consistency checks will quickly alert users to such a mistake.
The actual reason is a practical one, revealed by PAF's behaviour on entering short years.

Personal Ancestral File

FamilySearch PAF 5.2.18 from 2002 CE is the closest thing we have to a reference implementation. When you enter a year less than 1000 in PAF, PAF will pad it out to 4 digits by using leading zeroes on screen, and export those leading zeroes to GEDCOM.

Additional PAF behaviour reveals *why* FamilySearch does not like years of less than 3 digits. If you enter a year (without leading zeroes) of 31 or less, PAF does not add leading zeroes, but presents a pop-up message-box instead, with the complaint that “The date you typed is not standard”. That's a rather generic message, one that PAF even pops up for things that GEDCOM allows, but PAF does not support.

PAF only does this when the year is 31 or less, and that reveals the reason why. Not only are GEDCOM date formats quite flexible, users in different locales may also want to format dates differently. PAF simply does not know whether you mean `30 Mar 15` when you type `15 Mar 30`. The demand to use at least three digits for years is to avoid ambiguity between years and days.
Years must be at least three digits long to easily distinguish between days and years.

References

- FamilySearch GEDCOM Specifications
- Ancestral Question Version History: Version 3.0 Features

Alias ALIA

The ALIA record has a remarkable history.
FamilySearch GEDCOM 3.0 and 4.0 defined the ALIA (alias) record as way to record alternate names. FamilySearch GEDCOM 5.0 and later specify that alternate names must be recorded using multiple NAME records. At the same time, FamilySearch GEDCOM 5.0 defined a new ALIA record, namely as a way to record that another INDI record is possibly for the same person.

There was no reason to create confusion by calling this the ALIA record. In fact, there was no reason to create this record type at all, because GEDCOM already supports the <<ASSOCIATION_STRUCTURE>> (ASSO record).

The ASSO record should be used for all relationships that aren't direct family relationships. Possibly duplicate INDI records can be associated with each other using the ASSO record; `possible-duplicate` is a good ASSO.RELA value for that usage.

In practice, many products kept creating ALIA record to record alternate names. All versions of Ancestry.com Family Tree Maker kept creating ALIA records. Software MacKiev Family Tree Maker 2014.1 stopped creating ALIA records; it uses multiple NAME records as it should.

The new ALIA record, a link to another INDI record, has never been used. It only occurs in GEDCOM test files.
GEDCOM 5.5.1 Annotated Edition deprecated the INDI.ALIA record. GEDCOM 5.5.5 removed the INDI.ALIA record. The INDI.ALIA record is illegal now.

GEDCOM 5.5.1 Best Practice

GEDCOM Writer

- Do not use ALIA to record alternate names
- Use multiple NAME records to record alternate names
  - Export the main name first
  - Use additional NAME records to support alternate names
  - Use slashes to delimit the surname on every NAME record
- Do not use INDI.ALIA for linking to other INDI; consider it a deprecated feature
- You can use the ASSO record to link an INDI record to another INDI record

GEDCOM Reader

- Read multiple NAME records containing alternate names
- Assume the first NAME record is the main, preferred name
- Optionally, read INDI.ALIA and INDI.NAME.ALIA records containing alternate names
- If the INDI.ALIA line value appears to be a reference, do not import it, but warn that this legal usage is considered deprecated and not supported

References

- GEDCOM ALIA
- FTM 2017 GEDCOM

NOTE.SOUR.NOTE.SOUR...

The GEDCOM 5.5 errata sheet states that an optional reference to a SOUR record should be added to the NOTE_STRUCTURE definition, like this:

```
[
n NOTE @<XREF:NOTE>@ {1:1}
    +1 SOUR @<XREF:SOUR>@
|
n NOTE [<SUBMITTER_TEXT> | <NULL>] {1:1}
    +1 [CONC|CONT] <SUBMITTER_TEXT> {0:M}
    +1 SOUR @<XREF:SOUR>@
]
```

The GEDCOM 5.5.1 specification does *not* include the additional lines.

Loop

FamilySearch probably realised that this addition was a bad idea, because it allowed notes to have sources, which may have notes, which may sources, and so on, *ad infinitum*, and thus includes the possibility of a loop.

GEDCOM 5.5.1 allows source citations on top-level NOTE records, and allows notes on source records, but does not allow source citation on those SOUR.NOTE note records.